

Yang Ni

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🎓 Education

Tsinghua University

Ph.D. Candidate in Astronomy (Expected 2027), Advisor: Prof. Xuening Bai

2022 – Present

Beijing, China

Nanjing University

B.S. in Astronomy (Magna Cum Laude), Advisor: Prof. Bin Luo

2018 – 2022

Nanjing, China

🔍 Research Interests

- Early-Stage Evolution of Protoplanetary Disks; Planet Formation
- Evolution of Giant Molecular Clouds; Star Formation; Stellar Feedback

📄 Publications

(**F** for first author, **C** for co-author)

F2. *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks I. Direct Formation of Gas Giants via Disk Fragmentation,*

Y. Ni, H. Deng, X. Bai

ApJ, 995, 96

F1. *The Life Cycle of the Giant Molecular Clouds in Simulated Milky Way-mass Galaxies,*

Y. Ni, H. Li, M. Vogelsberger, F. Marinacci, L. Sales, P. Torrey

A&A, 699, A282

C3. *RIGEL: Feedback regulated cloud-scale star formation efficiency in a simulated dwarf galaxy merger,* Y. Deng, ..., **Y. Ni**, et al.

A&A, 704, A240

C2. *The THESAN-ZOOM project: Star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts,*

Z. Wang, ..., **Y. Ni**, et al.

MNRAS, 544, 2675

C1. *GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions,*

X.-Y. Wang, ..., **Y. Ni**, et al.

ApJ, 922, 237

👥 Conferences & Seminars

1. *Direct Formation of Gas Giants via Disk Fragmentation in 3D Radiation Hydrodynamic Simulations,*

Contributed Talk, Exoplanets & Planet Formation Workshop 2025

October 2025, Yunnan

2. *Early Protoplanetary Disk Evolution and Planet Formation,*

Contributed Talk, NSFC scientific meeting on CPD and satellite formation

May 2024, Beijing

3. *Radiation hydrodynamic simulations of self-gravitating young circumstellar disks,*

Poster, Exoplanets & Planet Formation Workshop 2023

December 2023, Beijing

4. *Temporal evolution of giant molecular clouds in galaxy formation simulations,*

Talk, Colloquium at Department of Astrophysics, Host: Prof. Lucio Mayer

November 2023, Universität Zürich

5. *Temporal evolution of giant molecular clouds in galaxy formation simulations*,

Contributed Talk, International Conference on Resolving Galaxy Ecosystems Across All Scales

December 2023, The Chinese University of Hong Kong

Technical Skills

- ✓ Developer of M1-radiative transfer module in GIZMO
- ✓ Experienced in (radiation-)hydrodynamic simulation with GIZMO
- ✓ IRAF/Pyraf; The Chandra Interactive Analysis of Observations (CIAO)
- ✓ Python: astropy, h5py, numpy, matplotlib, scipy, yt, ... ; C/C++; Mathematica; L^AT_EX

Honors & Awards

Selected:

National Astronomical Observatory of China (NAOC) Scholarship,	NAOC
People's Scholarship First Prize,	The Ministry of Education of the PRC
Outstanding Undergraduate,	Nanjing University
"Top-notch" First Prize,	Nanjing University
G. Zheng Scholarship for Oversea Study,	Zheng Foundation

References

(Current Advisor) **Prof. Xuening Bai**, Tsinghua University

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Prof. Hongping Deng, Shanghai Astronomical Observatory (SHAO)

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Prof. Hui Li, Tsinghua University

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Prof. Mark Vogelsberger, Massachusetts Institute of Technology

✉ mvogelsb@mit.edu

Prof. Bin Luo, Nanjing University

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